Simulation Report

tet

Pedigree based match probability results

Match probabilities

Type	Probability	Iterations	Model probabilities
Average pedigree probability[1]	5.8425E-02	60,000	5.9084E-02, 5.7971E-02, 5.8219E-02
Inside match probability[2]	• 1: 5.5897E-01 • 0: 4.4103E-01	60,000	5.5746E-01, 5.6098E-01, 5.5848E-01
Outside match probability[4]	5.8238E-02	60,000	5.8639E-02, 5.7782E-02, 5.8291E-02

There is a **55.9%** probability that at least one other (non-excluded) unknown individual in this pedigree matches the suspect haplotype.

Likelihood Ratio (LR) = It is **0.789** times more likely that the suspect is the only individual in this pedigree carrying the suspect haplotype, compared to the alternative that at least one other untyped individual in the pedigree also carries it.

There is a **5.82%** probability that a random individual just outside this pedigree matches the suspect haplotype.

Disclaimer: The match probabilities shown above are the result of a stochastic (randomized) simulation. Due to the nature of Monte Carlo sampling, results may vary slightly between independent simulation runs, even when using the same input parameters.

[1] The average pedigree probability is the probability of the current state of the pedigree, i.e. with all typed and untyped haplotypes.

[2] The inside match probability is the probability that at least one other (non-excluded) unknown individual in this pedigree matches the suspect haplotype.

[4] The outside match probability is the probability that an individual just outside this pedigree matches the suspect haplotype.

Per individual match probability

Individual	Match probability
2	49.1%
4	37.9%
6	5.96%
7	5.81%
5	4.1%

Performance Metrics

- Run time pedigree probability: 0:00:28.456814
- Run time proposal distribution: 0:00:45.346301
- Run time extended pedigree: 0:00:00.003098
- Run time outside match: 0:00:15.224575
- Total run time: 0:01:29.031307
- Total needed iterations: 180,000

Simulation metadata

Date: 2025-10-07 14:40

- **Two step mutation factor:** 0.03
- **Stability window:** 10000
- **Model validity threshold:** 0.02
- **Number of threads:** 1
- **Simulation name:** tet
- **Results path:** /media/nw_disk1/dion/repositories/MatchY/results/tet_20251007143910
- **Bias:** None
- **User name:** ste

Pedigree information

Nodes: 7
Edges (relationships): 6
Individuals:
1: known, excluded: no
2: unknown, excluded: no
3: suspect, excluded: no
4: unknown, excluded: no
5: unknown, excluded: no
6: unknown, excluded: no
7: unknown, excluded: no

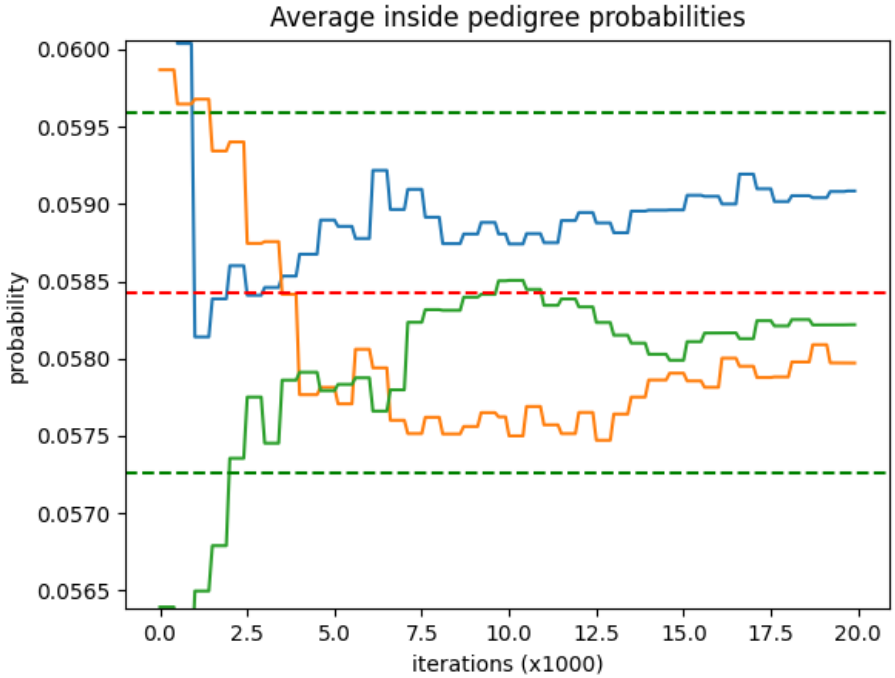
Known haplotypes

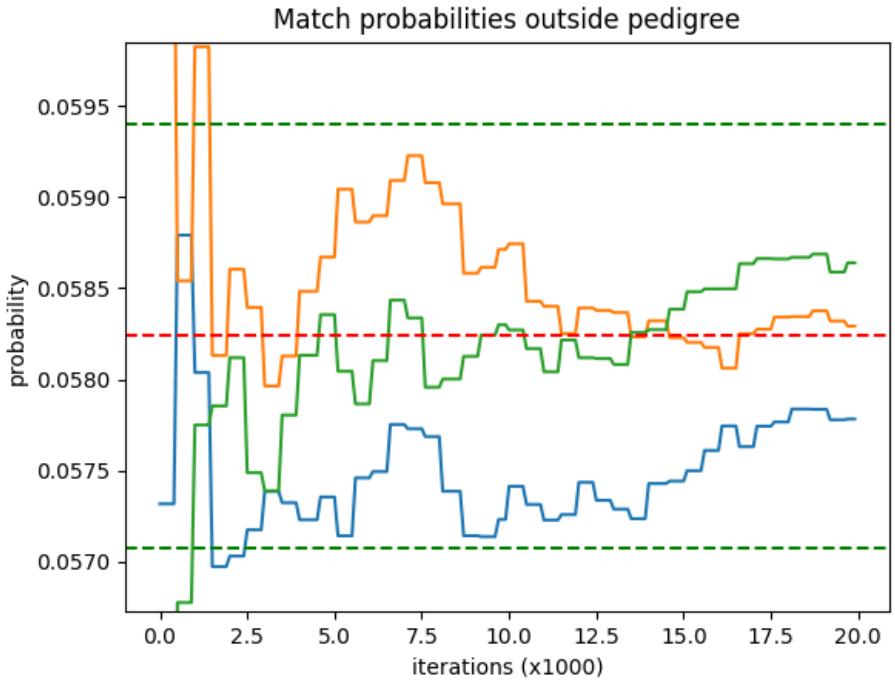
Marker	1	3
D1	10	10
D2	10, 11	10, 12

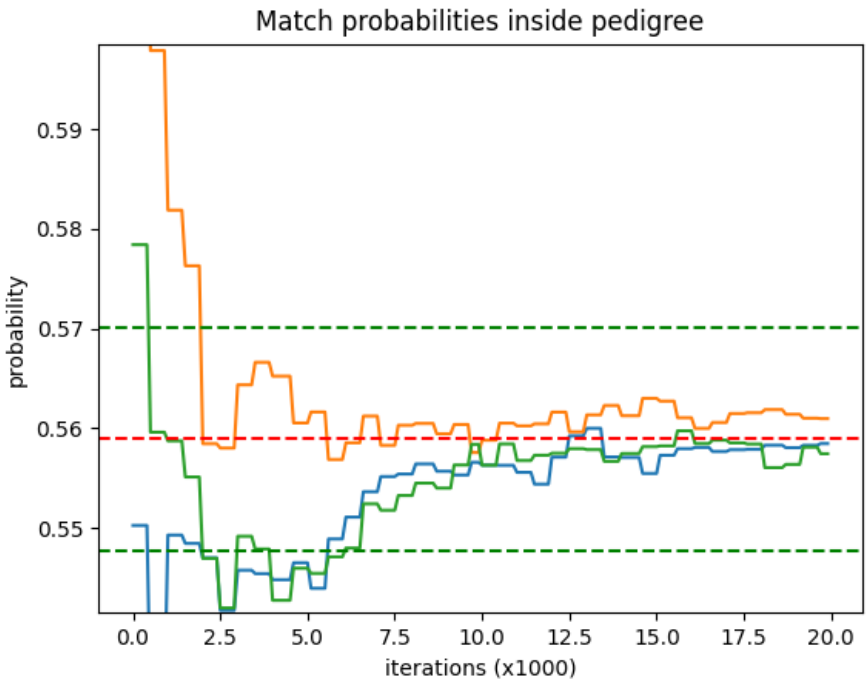
Marker set (marker: mutation rate)

- D1: 0.1
- D2: 0.2

Model validation plots

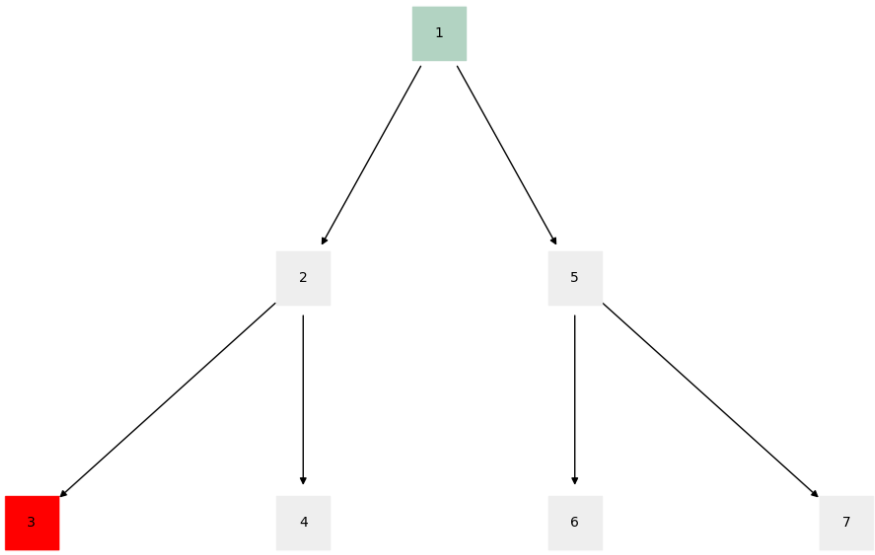

Average inside pedigree probabilities
probability
0.0600
0.0595
0.0590
0.0585
0.0580
0.0575
0.0570
0.0565
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
iterations (x1000)


Match probabilities outside pedigree
probability
0.0595
0.0590
0.0585
0.0580
0.0575
0.0570
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
iterations (x1000)

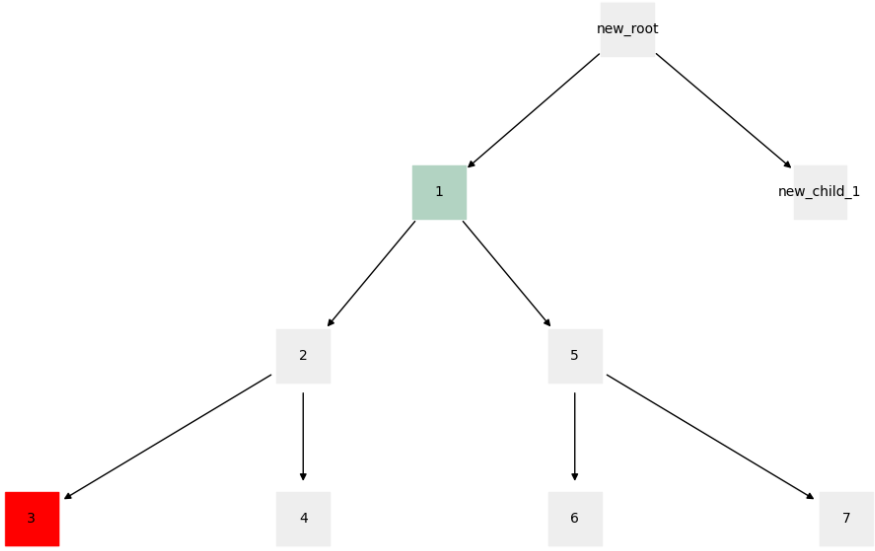


Pedigree configurations

Original Pedigree



Extended Pedigree



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